

1 Detailed properties of the three-mode chain

The three-mode system is the smallest chain with an unforced interior degree of freedom. It is therefore a useful setting in which to ask two questions that are difficult to isolate in long chains: whether the full stationary density can be reconstructed in the reduced five-dimensional state space, and which relaxation modes are actually visible in stationary trajectories. We treat these as separate inference problems. In particular, a successful trajectory fit is not taken as evidence for a stationary Fokker–Planck solution unless it first passes an equilibrium known-answer test.

1.1 Stationary state and the validation boundary

We use the reduced state

$$X = (I_1, I_2, I_3, \theta_1, \theta_3), \quad \theta_r = 2(\phi_r - \phi_2), \quad r \in \{1, 3\}, \quad (1)$$

and the energy

$$E(X) = \frac{1}{2}M^2 - \frac{1}{4} \sum_{j=1}^3 I_j^2 + I_1 I_2 \cos \theta_1 + I_2 I_3 \cos \theta_3, \quad M = I_1 + I_2 + I_3. \quad (2)$$

At equal bath temperatures, the exact invariant density with respect to $dI_1 dI_2 dI_3 d\theta_1 d\theta_3$ is proportional to $\exp[-E(X)/T]$. This Gibbs state provides a stronger validation target than agreement with histograms alone, because it tests both the density and the stationary adjoint equation.

The controlled trajectories were generated using double precision, fixed timesteps, standard trigonometric functions, implicit-midpoint Hamiltonian updates, Cartesian bath increments, and no positive-action projection. For each bath condition and timestep, complete trajectories—not individual snapshots—were assigned to 32 training, 16 validation, and 16 blind-test streams. The minimum effective sample sizes on the fine-timestep test splits were 1.16×10^4 out of equilibrium and 1.39×10^4 at equilibrium, above the preregistered threshold of 5×10^3 .

We tested a normalized conditional mixture in the logarithmic actions. The model combines a Gaussian mixture for $u_j = \log I_j$ with conditional periodic von Mises factors for (θ_1, θ_3) , and includes the Jacobian $\rho(I, \theta) = p(u, \theta)/(I_1 I_2 I_3)$. Before fitting, we froze six model families ($K = 8$ or 16, two network widths, and two stationary-equation penalties) and three seeds per family. Selection was allowed only if all three seeds reproduced the equilibrium Gibbs slope and satisfied pointwise $L^\dagger \rho = 0$ residual gates.

Figure 1 shows the complete equilibrium validation result. Although several likelihood fits reproduce the Gibbs log-density slope approximately, none solves the stationary equation at the declared accuracy. Across the 18 fits, the median of $|L^\dagger \rho / \rho|$ is 1.13–2.62 (gate 0.20), and its 90th percentile is 3.71–9.46 (gate 1.00). The centered Gibbs log-density RMSE is 0.302–0.595. Hence no architecture is admissible, even before opening the equilibrium blind split.

This is a negative but decisive result: the tested neural family does not support a numerical-solution claim for the five-dimensional stationary Fokker–Planck equation. In accordance with the frozen protocol, we did not train the corresponding NESS density. Consequently, density-derived claims about low-probability stationary geometry are not used below.

1.2 What is established about stabilization

The failure above concerns density reconstruction, not the existence of a stationary trajectory ensemble. Independent audits of the reduced generator give maximum drift and covariance discrepancies of 2.3×10^{-13} and 2.9×10^{-11} , respectively, while the Gibbs adjoint residual is 1.5×10^{-14} in RMS on prespecified interior points. Moreover, a simultaneous equilibrium calibration of 18 regular weak identities yields a family-wise threshold of 3.748 standard errors, with no failures on the equilibrium or driven held-out trajectories. These tests support the use of the controlled trajectories for stationary expectations and dynamical correlations.

They do not, however, identify a quantitative high- versus low-energy stabilization mechanism in the full state space. Such an interpretation would require either a stationary density that passes the Gibbs and adjoint tests above, or a separate set of directly estimable conditional balance relations with their own held-out controls. We therefore leave the detailed density geometry unresolved instead of inferring it from an unvalidated neural solution.

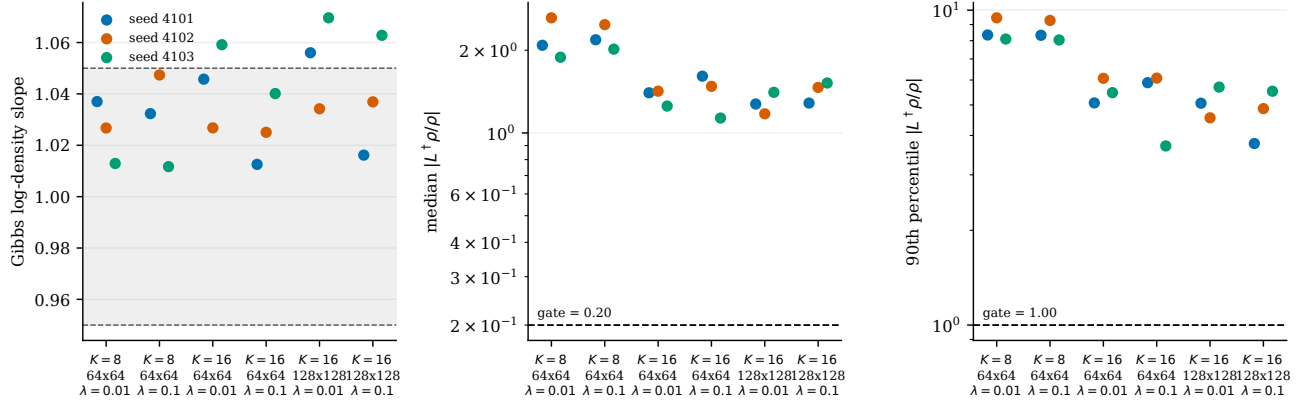


Figure 1: Equilibrium known-answer calibration of the normalized stationary- density candidates. Each symbol is one independent training seed. The shaded region in the left panel is the validation slope gate; dashed lines in the residual panels are the preregistered pointwise Fokker–Planck gates. All candidates violate the residual gates.

1.3 Held-out relaxation modes of the generator

The spectral analysis is independent of the density reconstruction. We fit extended dynamic mode decomposition (EDMD) operators on 32 trajectories, selected dictionaries and regularization on 16 validation trajectories, and froze the eigenvectors before a single evaluation on 16 test trajectories. The dictionary includes raw endpoint-action differences and their prescribed cross terms with the angular basis, so that it can represent the known exchange-odd oscillation. The slow real candidate is selected at lag $\tau = 0.5$, whereas the complex candidate is selected at $\tau = 0.05$ to avoid principal-log aliasing.

The selected slow real eigenvalues lie between -0.91 and -0.95 across bath conditions and timesteps. On the blind trajectories, their observable autocorrelations yield rates -0.884 and -0.909 in the driven system, and -0.951 and -0.826 at equilibrium, for timesteps 10^{-3} and 2.5×10^{-4} , respectively. As shown in Fig. 2, the corresponding bootstrap intervals overlap between timesteps and the held-out Koopman residuals do not grow by more than the preregistered 25% tolerance.

The targeted complex mode is recovered as well. At the fine timestep, the blind frequencies are 6.281 with 95% interval $[5.388, 6.862]$ in the driven case and 5.088 with interval $[4.487, 5.558]$ at equilibrium. These intervals contain the independent observable-level estimates 5.500 and 5.264. The coarse-timestep intervals exclude zero only narrowly and are much broader, so we do not use them for a precision claim.

Finally, the slow eigenfunction has a reproducible observable content (Fig. 3). Its absolute modal weight is 0.47–0.50 in I_2 , 0.28–0.30 in $I_1 + I_3$, 0.20–0.22 in the cosine sum, and 0.106–0.124 in $\cos \theta_3$. This ordering explains why different raw observables approach their asymptotic correlation rates with different visibility, without requiring a different leading mode for each observable.

The available evidence therefore resolves a held-out, EDMD-visible slow mode and a faster damped oscillation. It does not establish that the real mode is the unique spectral gap of the full generator: the preregistered common-autocorrelation-plateau gate still fails. We retain this distinction in all subsequent claims.

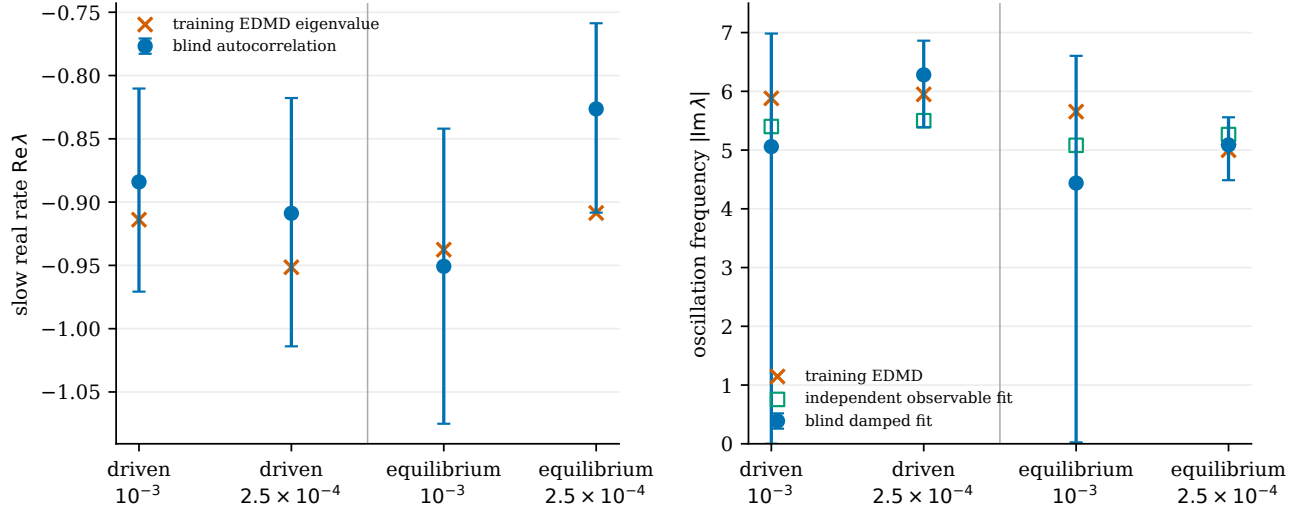


Figure 2: Held-out validation of the real and oscillatory EDMD modes. Left: selected training eigenvalues and rates measured from the frozen eigenfunctions on blind trajectories. Right: EDMD frequencies, independent observable-level fits, and blind damped-oscillation fits. The coarse-step blind intervals are broad; the fine-step fits provide the informative frequency comparison.

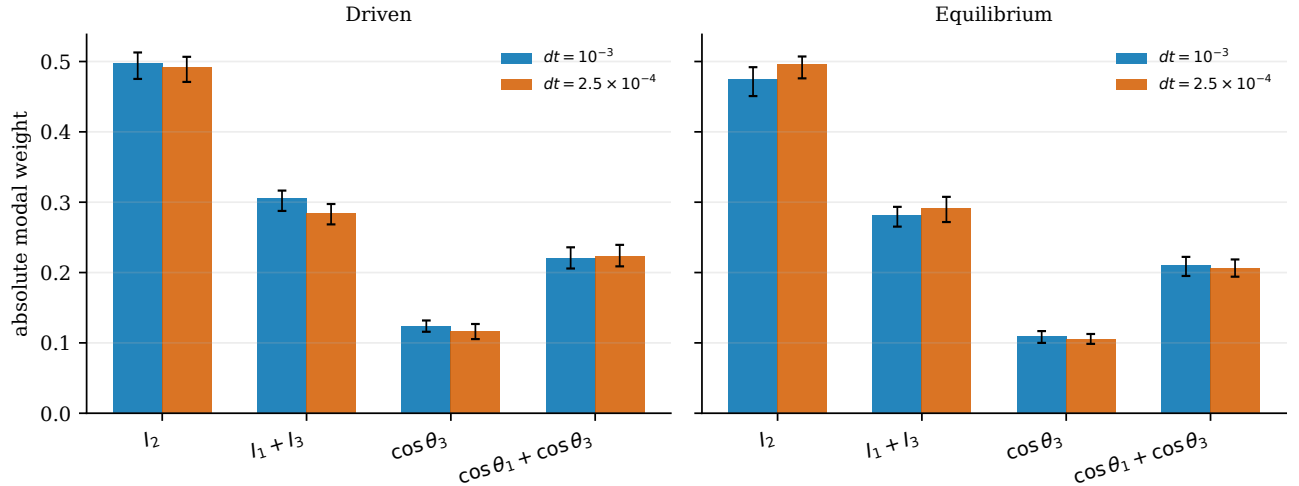


Figure 3: Absolute weights of four observables in the frozen slow real mode, evaluated on blind trajectories. Error bars are 95% whole-trajectory bootstrap intervals. The ordering is stable across bath conditions and timesteps.